

P r o b l e m S e t # 1

1. Renting Cloud Storage Capacity

You are getting ready to launch a really cool new app. You are hoping it will go viral, giving you a surge of customers in a very short period of time. But who knows, it might flop.

You have everything in place except for cloud storage capacity, which you still need to rent. Demand for your app will either be “High” or “Low.” High demand means you will have 20 million users. Low demand means only half as many. Your best guess is that the probability of “High” demand is two-thirds. Each user will pay \$1 for the app. But you absolutely will not accept any customer unless you have sufficient cloud storage capacity to serve that customer. All of your costs other than the cloud storage capacity have already been incurred, so we can ignore those for your current decision. You are risk neutral for this decision.

You are considering renting your cloud storage capacity from the leader in the market, Nile. Nile rents cloud storage capacity in standardized units, each of which is sufficient to serve 10 million of your users. Nile has a two-tier pricing structure: “advance commitment” units cost \$4 million each, and “on-demand” units cost \$5.5 million each. The timing is such that you would need to rent any “advance commitment” units before you know whether demand is “High” or “Low.” In contrast, “on-demand” units from Nile are available immediately upon placing an order, so you can order them after you learn about demand. If you order an “advance commitment” unit from Nile and don’t use it, you won’t get any refund.

- a) Draw and solve the decision tree you face.
- b) You really are not sure how likely it is that demand will be “High.” Denote by P the probability that demand is “High.” At what level of P does your decision change?

We now consider an expanded decision problem, where you also have a choice of using another cloud storage vendor, Volga. For technical reasons, you cannot split your business between the two vendors; you must pick one or the other. (For the remainder of the problem, assume again that the probability of “High” demand is two-thirds.)

Volga rents the same standardized storage units, but they have a simpler pricing structure: all of Volga’s storage units are available immediately upon placing an order, for \$4.5 million per unit.

- c) Draw and solve the decision tree you face.

2. Research Project

AK Steel Holding Corporation is a producer of flat-rolled carbon, stainless and electrical steels and tubular products through its wholly owned subsidiary, AK Steel Corporation. The recent decline in the demand for steel prompted AK executives to look for new ways to cut costs, so they are considering a research project to improve production of rolled steel. The research involves three distinct steps, each of which must be successfully completed before the firm can implement the cost-saving new production process. If the research is completed successfully, it will save the firm \$4 million. Unfortunately, there is a chance that one or more of the research steps might fail, in which case the project is worthless. The three steps are done sequentially so that the firm knows whether one step was successful before it has to invest in the next step. Each step has a 0.8 probability of success and each step costs \$500,000. The risks of failure in the three steps are uncorrelated with one another. Assume that AK Steel is a risk neutral company (*i.e.*, that it maximizes expected values). In case you are worried about such things, the interest rate is zero.

- a) Draw the decision tree for the firm.
- b) If the firm proceeds with this project, what is the probability that it will succeed in implementing the new production process?
- c) If the research were costless, what would be the firm's expected gain from it before the project began?
- d) Should the firm begin the research, given that each step costs \$500,000?
- e) Once the research has begun, should the firm quit at any point even if it has had no failures? Should it ever continue the research even if it has had a failure?

After the firm has successfully completed steps one and two, it discovers an alternate production process that would cost \$150,000 and would lower production costs by \$1,000,000 with certainty. This process, however, is a substitute for the three-step cost-saving process; they cannot be used simultaneously. Furthermore, to have this process available, the firm must spend the \$150,000 before it knows if it will successfully complete step three of the three-step research project.

- f) Draw the decision tree faced by the firm after it has successfully completed steps one and two, including the possibility of pursuing this alternate production process.
- g) If the firm continues the three-step project, what is the chance it would get any value at all from also developing the alternate production process?
- h) If developing the alternate production process were costless and if the firm continues the three-step project, what is the expected value that it would get from having the alternate production process available (viewed at the beginning of research step 3)?

- i) Should the firm in this situation:
 - a) Pursue only the third step of the three-step project
 - b) Pursue only alternate production process
 - c) Pursue both the third step of the three-step project and the alternate process
- j) If the firm had known of the alternate production process before it began the three-step research project, what should it have done?

3. Upgrading Production Facilities

Consider a firm that produces widgets and expects next year's profits to be \$940,000. It is considering upgrading the production facility, which it owns, to replace the production method with a new and more efficient method. The firm currently owns a building, half of which it uses for production and the other half it rents out at a monthly rate of \$50,000. The upgrade will require using the whole building. Also, the firm would have to raise \$4 million to pay for new equipment. They can currently secure a commercial loan with an annual interest rate 6%. Furthermore, the firm expects to spend \$80,000 during the first year of operation to maintain the market value of the equipment. The new process implies added costs of material, labor, and utilities during the first year of operation. The additional costs are \$1 million for materials, \$800,000 for labor and \$280,000 for utilities above and beyond what they would be under the old production method. How much higher would the anticipated yearly revenue have to be with the new production method in order for this change to be profitable in the first year?

4. Internal vs. External Pricing

Qlink has strong patent protection on a particular type of computer chip. It can produce an essentially infinite number of these chips at a constant cost of \$50 a chip. Qlink uses the chip in internal production of handsets. It also sells the chip to other manufacturers in the open market. Does Qlink necessarily want to charge the same price to both internal and external customers? Explain.

5. Under New Management

Margie Sharp has been installed as CEO of SC Enterprises that currently operates out of a single building, built in 1998 for \$300 million in cash. Given its out-of-the-way location, it is believed that no one would buy it were the building put up for sale. SC Enterprises makes four toy lines: puzzles, teddy bears, dolls, and toy guns. Each toy line has its own assembly line and employees who work on that assembly line. The lines, however, all go through the same packaging and shipping department. Each type of toy takes the same time and uses the same amount of materials to be processed through packaging and shipping (e.g., a puzzle and a teddy bear both take the same time to be processed and use the same amount of packaging materials). The *previous management's* accounting for these toy lines is shown below.

Annual Accounting at SC Enterprises

[All figures in millions. Numbers in parentheses represent negative amounts.]

	Puzzles	Teddy Bears	Dolls	Toy Guns	SC Enterprises
Number sold	1	1	1	2	5
Revenue	\$9.90	\$20.00	\$20.00	\$11.80	\$61.70
Direct labor-hours	1	1	1	1	4
(@ \$4 per hour)	\$4.00	\$4.00	\$4.00	\$4.00	\$16.00
Other direct costs	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
Overhead*	\$7.50	\$7.50	\$7.50	\$7.50	\$30.00
Profit	(\$1.60)	\$8.50	\$8.50	\$0.30	\$15.70

* Overhead includes total expenditures on packaging and shipping (\$20M) and the cost of the building (\$10M, calculated using 30-year straight-line depreciation). Overhead is allocated on the basis of *direct* labor-hours (e.g., if a line uses 25% of all direct labor, then \$7.5 million [= 0.25·30] of overhead is allocated to that line).

- (a) What is the annual economic cost of operation for SC Enterprises?
- (b) What is the annual economic cost of operating each of the four toy lines?
- (c) Which, if any, of the toy lines should Ms. Sharp eliminate (i.e., shut down)?